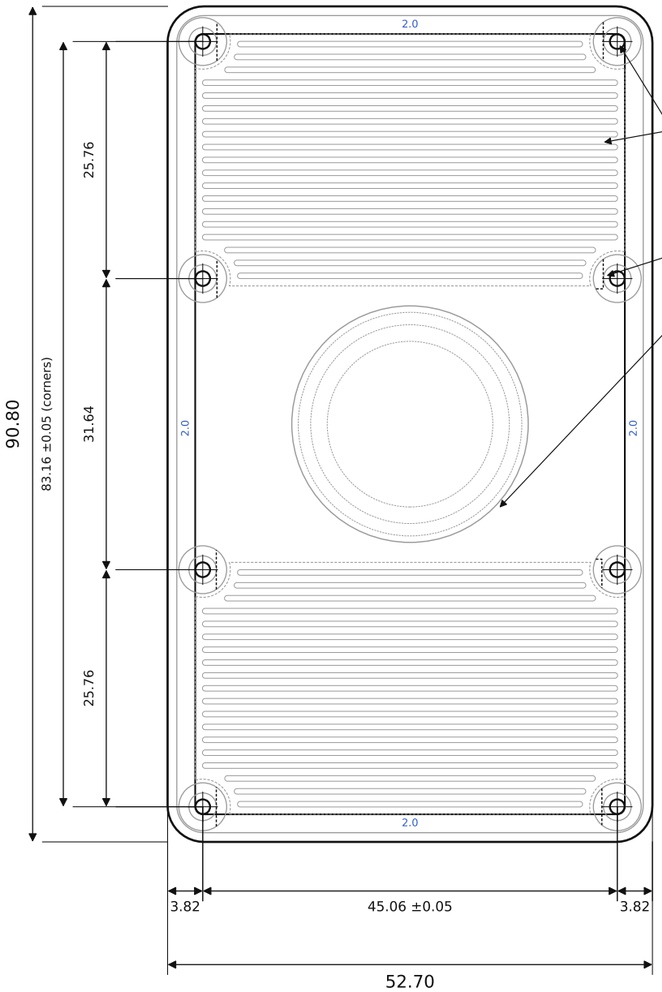
BACK FACE (outer) SCALE 1.5:1



REEDED FIELDS ×2 (REF — REAR ART BLOCK)

8× BOSS FLAT (HIDDEN — CAVITY SIDE): EACH Ø5.20 BOSS CUT BACK 0.25–0.75 CLEAR OF THE PART IT WOULD FOUL (SC1 SC2 SC3 SC4), FULL DEPTH; FLATS 1.47–1.68 FROM THE HOLE AXIS (THREAD KEEP-OUT r1.30 UNTOUCHED). MODELED — STEP GOVERNS.

MEDALLION (REF — REAR ART BLOCK)

POS	Ø0.10 (M)	A	B

8× M2 THREAD, THRU

Ø1.6 TAP DRILL • TAP FROM BACK FACE

Ø3 SPOTFACE, BACK (depth per Detail B)

C3 = hole-pattern pitch ±0.05   C4 = M2 thread

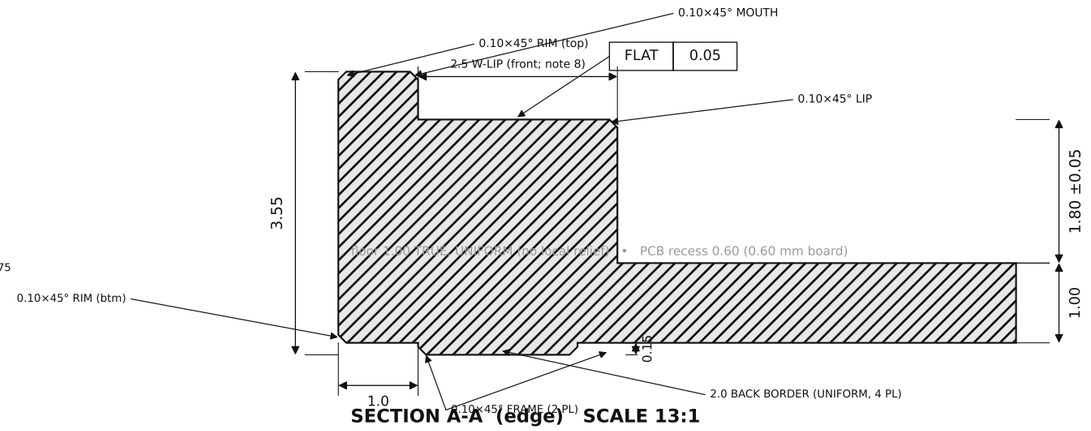
SECTION A-A → edge profile *DETAIL B → corner boss*

REAR ART — MACHINED-IN GEOMETRY (REFERENCE; STEP GOVERNS)

REEDED FIN FIELDS ×2 (PLAN, dashed zones): 19 RIBS/FIELD, 0.60 WIDE ON 1.399 PITCH; GROOVES Ø0.6 END-MILL ONLY (0.80 WIDE), VALLEY 0.60 BELOW ART FIELD (0.40 WEB); RIB TOPS +0.10 — 0.05 UNDER BEARING PLANE.

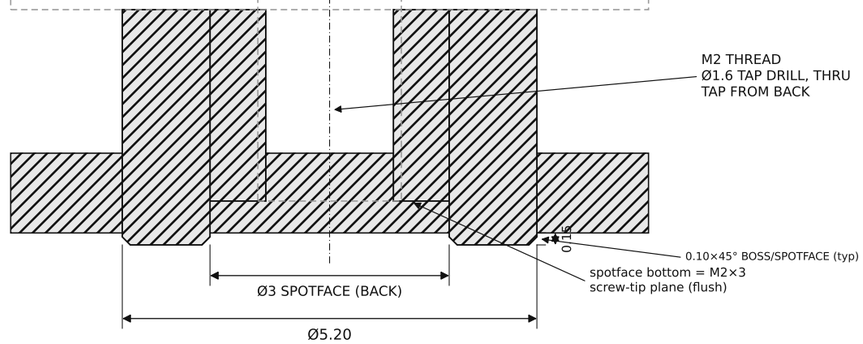
MEDALLION (PLAN, circles): Ø25.7 COIN SUNK 0.45 BELOW ART FIELD (Ø0.4 END MILL), COUNTERS/REST 0.25 (Ø0.3 REST PASS); CRESTS (RING TEXT, RIM, HOOP, MONOGRAM, SERIAL) LEFT AT THE FRAME'S BEARING PLANE (−0.15), VERTICAL WALLS, NO DETACHED ISLAND UNDER 0.53. DO NOT EDGE-BREAK CREST EDGES. SERIAL ("No 001") IS PER-UNIT VARIABLE TEXT.

OP ORDER: ALL BACK-FACE FEATURES BEFORE CAVITY HOLLOW (SOLID-STOCK RULE). STRESS-RELIEVED STOCK — STATE CONDITION ON QUOTE.



SECTION A-A (edge) SCALE 13:1

C1 = cavity depth   C2 = PCB-rest-plane flatness   (back face down; PCB drops in from top)
PCB + M2×3 BRASS SCREW — REF, NOT THIS PART



DETAIL B (corner boss) 13:1

CRITICAL DIMENSIONS — C1 & C3 TOLERANCED ±0.05 (MARKED ON VIEWS); ALL OTHER FEATURES PER ISO 2768-1 (MEDIUM)

C1	Cavity depth (boss-top plane → cavity floor)	1.80 ±0.05
C2	PCB-rest plane flatness (lip + 8 bosses)	FLAT 0.05
C3	8× mounting holes (x 45.06; y rows 2.87/28.63/60.27/86.03)	±0.05 (linear)
C4	Mounting holes — tapped, thru, from back	M2 / Ø1.6 drill

NOTES

1. MATERIAL: TITANIUM Gr5 (TC4) = Ti-6Al-4V GRADE 5. [Ti Gr5 (reference; brass or 6061 acceptable -- non-magnetic, machinable)]
2. FINISH: BEAD-BLAST MATTE, ALL OVER. NO MARKING SERVICE — THE REAR ART (REEDED FIELDS + MEDALLION) IS MACHINED-IN, IN THE STEP (SEE REAR ART BLOCK). BEARING PLANE (FRAME + MEDALLION CRESTS) HAND-LAPPED BY CUSTOMER AFTER BLAST — NO SHOP ACTION.
3. GENERAL TOLERANCE PER ISO 2768-1 (MEDIUM). C1-C4 TOLERANCED AS LISTED; C1 & C3 = ±0.05. DATUMS: A = LEFT EDGE, B = BOTTOM EDGE, C = PCB-REST PLANE.
4. BRACE REGISTRATION: THE RESIN BRACE (1 PIECE, ITS FOOTPRINT COMPUTED FROM THE BOARD) REGISTERS TO THIS SHELL BY FITMENT - ITS OUTLINE, 0.05 OFF THE CAVITY WALLS WHERE IT REACHES THEM, + THE COMPONENT POCKETS + THE BOARD IN ITS RECESS (RECESS 50.70 x 88.80 FOR A 50.80 x 88.90 BOARD: 0.05 INTERFERENCE PER FLAT AS MODELLED). NO LOCATOR PILLARS; THE CAVITY FLOOR IS A FULL 1.00 EVERYWHERE.
5. EDGE BREAKS - ALL 0.10x45°, FELT NOT SEEN. MODELED, CALLED OUT ON SEC A-A / DETAIL B: RIM = outer rim (top & bottom); MOUTH = recess mouth (around PCB); FRAME = proud back-frame bottom edges; BOSS = boss + spotface bottom edges (Detail B). LIP = inner (cavity-side) lip edge: BREAK EVERY LIP EDGE ON THE PART, WHETHER OR NOT THE MODEL SHOWS IT. BREAK ALL OTHER EXPOSED EDGES 0.10x45°. CONCAVE JUNCTION CORNERS LEFT SHARP IN THE MODEL: FINISH WITH A Ø2.0 mm TOOL = R1.0 AS-MILLED.
6. FLOOR IS A TRUE 1.00, UNIFORM ACROSS THE CAVITY, AT THE ~1.0 Ti MIN-WALL GUIDANCE SO IT ALSO CLEARS ALUMINIUM / COPPER / STAINLESS. CAVITY 1.80 ±0.05 -> 1.75 WORST-CASE, MINUS WS17 1.70 MAX (DATASHEET CASE WS17 HEIGHT) = 0.05 NON-CONTACT; THE BRACE + SOLAR-CELL SANDWICHES CARRY THE BOARD.
7. NO U7 RELIEF POCKET. U7 IS THE 0.90 mm MB85RC512TY DFN-8, WELL UNDER THE 1.80 CAVITY, SO THE FLOOR IS UNIFORM 1.00. (A POCKET EXISTED FOR A 1.75 SOIC-8 THE v4 BOARD DOES NOT CARRY.)
8. FRONT SUPPORT LIP IS CUT PER BAND TO CLEAR B-SIDE PARTS (SECTION A-A: THE WEST LIP AT FULL WIDTH). WIDTHS FROM THE BOARD EDGE, POSITIONS IN BOARD x/y, LIMITER IN BRACKETS: WEST 2.50; 2.32 y45.5–48.5 (R16), 2.25 y48.5–49.5 (C24). EAST 1.25 FULL LENGTH (NFC COIL, JP1, TP1). SOUTH 2.00; NONE x4.3–22.9 (SC1), x27.8–46.3 (SC2). NORTH 2.00; NONE x4.4–22.9 (SC3), x27.9–46.4 (SC4). A BAND STOPS 0.30 SHORT OF ITS PART, OR 1.30 WHERE THE Ø2.0 TOOL COULD NOT CLEAR IT, AND 1.00 SHORT OF NFC COIL COPPER (GROUNDED METAL OVER THE COIL WOULD DETUNE IT). STEPS BLEND AT R1.00. THE EXTERIOR BACK BORDER (PLAN) IS INDEPENDENT AND UNIFORM 2.0 ON ALL 4 SIDES.
9. NO INTERNAL RIBS OR SUPPORT POSTS: THE RESIN BRACE (SEPARATE PART) CARRIES CENTER SUPPORT. A PCB LAYOUT CHANGE = A BRACE REPRINT, NOT A SHELL RE-MACHINE.
10. PCB RECESS = 0.60 DEEP, RECEIVES THE 0.60 mm BOARD (RECESS 50.70 x 88.80 FOR A 50.80 x 88.90 BOARD: 0.05 INTERFERENCE PER FLAT AS MODELLED). 3D STEP GOVERNS ALL GEOMETRY; STL NOT FOR CNC.
11. PART = BACK-SHELL ONLY. PCB, RESIN BRACE, AND 8× M2×3 BRASS SLOTTED-CHEESE SCREWS (HEAD Ø3.8) SUPPLIED SEPARATELY.

GENERAL TOLERANCES (UNLESS OTHERWISE NOTED, mm)

LINEAR & HOLE-POSITION DIMS	**ISO 2768-m**
CAVITY DEPTH C1 (SECTION A-A)	**1.80 ±0.05**
HOLE-PATTERN PITCH C3 (8 HOLES)	**x 45.06 / y 25.76-31.64-25.76 ±0.05**
PCB-REST FLATNESS C2	**FLAT 0.05**

SOLAR-GLOW DRH v4.0 — Ti BACK-SHELL (0.6mm-BOARD DUMB BOX)

DWG solar-glow-drh-v4_0-backshell-0p6b-brace	REV v4.0
MATERIAL Ti Gr5 (TC4)	FINISH BEAD-BLAST + LAP (CUSTOMER)
UNITS mm	SCALE AS NOTED
BBOX 52.70 x 90.80 x 3.55 3rd-angle	SHEET 1/1